This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Abandoned Object Detection and Classification using Deep Embedded Vision

Arbab Muhammad Qasim[1], Naveed Abbas[2], Amjid Ali[3], Bandar Ali Al-Rami Al-Ghamdi[4]

[1,2,3]Department of Computer Science, Islamia College University Peshawar, KPK, PAKISTAN
4 Faculty of Computer Studies, Arab Open University, Riyadh, KSA.

Corresponding author: Naveed Abbas (e-mail: naveed.abbas@icp.edu.pk)

This work was supported by Arab Open University, Riyadh Saudi Arabia, Faculty of Computer Studies under research grant no AOURG-2023-010. The authors are thankful for the support.

ABSTRACT One indispensable element within security systems deployed at public venues such as airports, bus stops, train stations, and marketplaces is video surveillance. The evolution of more robust and efficient automated technological solutions for video surveillance is imperative. In light of the escalating global threat of terrorist attacks in recent years, any unattended object in public areas is treated as potentially suspicious. Ensuring the protection of individuals in these public spaces necessitates the implementation of safety measures. The intricacies of surveillance recordings introduce challenges when it comes to identifying abandoned or removed objects, owing to factors like occlusion, abrupt lighting changes, and other variables. This paper proposes a novel two-stage method for identifying and locating stationary objects in public settings. The first stage uses a sequential model to capture temporal features and detect potential abandoned objects within the monitored area. When the sequential model detects such an object, it triggers a subsequent phase. The second stage uses YOLOv8l model to precisely locate the detected objects. YOLOv8l is renowned for its ability to accurately pinpoint object locations within the surveillance scene. The proposed method achieves remarkable accuracy rates of 99.20% and 99.70% on combined PETS 2006 and ABODA datasets, respectively, effectively localizing the target object. This achievement not only underscores the model's precision in accurately pinpointing the object's position within the given context but also establishes its superiority over other existing models. By integrating these two stages, our method provides an effective solution for enhancing the detection of abandoned objects in public spaces, contributing to improved security and safety measures.

INDEX TERMS Abandoned object localization, Stationary object detection, Embedded vision, abandoned object, video-surveillance

I. INTRODUCTION

In the various application of computer vision, Video surveillance is attracting the attention of researchers, and actively search for detecting and tracking the object in the videos [1]. In real time application, intelligent video surveillance systems are drastically developed to automate the surveillance [2]. The smart video surveillance system autonomously identifies specific occurrences like trespassing, lingering, and abandoned objects [3-6]. Researchers have explored object detection in videos by approaching it as the recognition of objects within each individual frame, essentially treating each frame as a standalone image [7]. Video surveillance mainly consists of object detection and tracking such as automotive driving, and intelligent robotic technology [8]. In the realm of video surveillance, there is a focus on dynamic environments to track cars and various real-world objects [9]. In computer vision applications, object tracking and object detection in video surveillance progress hand in hand [10]. Object detection involves classifying and locating objects or instances of interest within a suspicious frame, whereas object tracking is the process of recognizing the trajectory across consecutive frames [11]. Conversely, the static detection and segmentation of objects in videos remain a challenging and actively researched area [10]. Object segmentation and static object analysis involve the identification, tracking, and assessment of an object's presence [12]. The most recent advancement in video surveillance technology allows it to automatically identify abandoned objects in public areas and illegally parked cars in traffic monitoring systems [13]. Detection of an abandoned object in video surveillance is challenging and essentials for maintaining safety [14]. Manually processing such huge amount data seems impossible and time consumption, An additional measure for public safety is the automatic identification of abandoned items [15].

An abandoned item is an object that has been left behind by its owner and has not been reclaimed within a predetermined period of time [5]. Mostly existing abandoned objects detecting algorithms utilized background model for extracting foreground information [16]. The background model serves as an effective method for extracting

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

foreground information from images. However, a notable challenge arises over time, wherein the distinction between the foreground and background becomes less discernible. In other words, as the processing continues, the initial clarity between the foreground and background diminishes, as highlighted by the gradual merging of these elements [17]. This phenomenon creates a hindrance for algorithms that heavily rely on foreground information to detect and track target objects [18]. Additionally, these models are susceptible to variations in lighting, which can alter the image's shape and lead to unstable model outputs. These effects can notably elevate the false alarm rate, resulting in the subpar performance of the surveillance system [19]. One of core elements of these systems is object detection and tracking, which watch the target over time [20]. Furthermore, the use of video surveillance cameras to identify suspicious situations has significantly increased in the past few years [21]. With recent progress in object detection and facial classification, video surveillance systems incorporating both object detection and facial recognition have become more prevalent [22]. Motion detection-based methods are used for abandoned object detection in surveillance system [23], the method consists of background subtraction techniques, followed by optical flow analysis techniques and temporal differences technique [23]. Detecting intentionally discarded or abandoned objects within a scene poses a significant challenge for object detection techniques. Unattended object detection, which identifies unattended items in a series of video frames, aims to address this issue [24].

One of the domains that Artificial Neural Networks (ANNs) have been successfully applied to is computer vision, which is the field of study that enables machines to understand and interpret visual information, such as images, and videos. All ANNs possess a shared capability to extract and learn high-level features from visual information, enhancing machines' ability to comprehend and interpret visual data more effectively. Abandoned object detection is a challenging task which not only machines require to locate and segment objects in complex scenes, but also classify them. Recent study [1] has used ANN for abandoned object detection in outdoor environments, which detects hand luggage using a deep learning-based detection method. However, their method only focuses on the hand luggage object, and does not consider other types of objects that may be abandoned, such as backpacks, and boxes.

In this paper we propose a novel method for abandoned object detection that can handle both suspicious and non-suspicious scenes. Our method consists scene classification module (SCM) and an object detection module (ODM) which will be discussed in more detail in methodology section. Our main contributions are as follows:

We develop a novel two-stage method for abandoned object detection that can adapt to different types of scenes and reduce false positives.

Our proposed model achieves an exceptional level of precision, with an impressive accuracy rate of 99.20% for classifier and 99.70% for the localizer, setting a new standard in abandoned object detection and surpassing existing methods.

Our proposed method outperforms all other existing methods on two public datasets, ABODA and PETS 2006.

The paper is structured to provide a clear progression of the research. It starts with a review of related work, offering context and identifying research gaps. Following this, the proposed methodology introduces the innovative two-stage model for abandoned object detection, addressing surveillance system challenges. The experimental evaluation section is divided into three sub-sections. First, the "Experimental Setup" outlines the tools and data sources used. Then, the "Experimental Results" demonstrate the model's performance with thorough analysis and evidence of its superiority over existing approaches. This structure ensures a logical flow from background research to the proposed solution and empirical support for the research's contributions

II. RELATED WORK

Several methods were used in this study, including object tracking, object identification, and object classification. Diverse techniques were applied to delineate the background and foreground regions of stationary objects over a certain duration. Many prior investigations on abandoned object detection have focused on the analysis of foreground information derived from one or multiple background models. This analysis is conducted to discern the differentiation between stationary and dynamically moving objects. Subsequently, the stationary objects are tracked over a specified duration to ascertain whether they exhibit characteristics indicative of abandonment. In their work, Fan et al. [25] achieved a reduction in the false alarm rate through the utilization of a single background model and a finite state machine (FSM). They accomplished this by modeling objects that exhibit temporary static behavior, such as a car that briefly halts and subsequently resumes movement. Their investigation also encompassed the concept of "healed" objects, referring to those objects that have already assimilated into the background. However, it's noteworthy that their study did not address the issue of illumination changes. Omrani et al. [26] introduced a system using stereovision for detecting and tracking objects in maritime environments. An autonomous surface vehicle (ASV) with a stereo camera was used to test the system. In the first stage, the ASV approached stationary objects in order to identify both static and dynamic things. After that, the ASV tracked a target boat using RTK-GPS to determine its absolute and relative positions. To differentiate between moving and static objects, image processing and stereo vision techniques were applied.

Narwal and Mishra [27] presented a system designed specifically for real-time unattended baggage detection, using frame captures from video surveillance cameras. The system encompasses several phases. It initiates with background subtraction, followed by subsequent steps that leverage the background-subtracted frames for recognizing

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

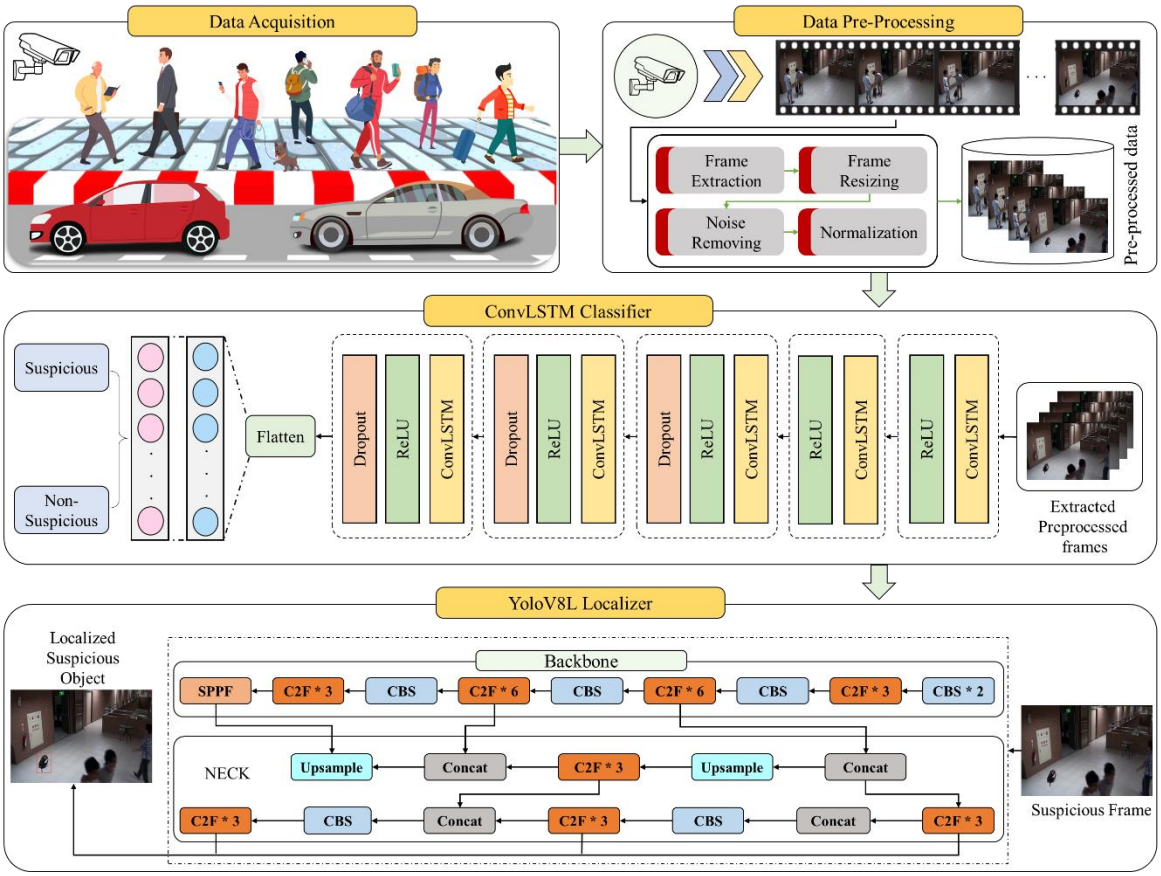


FIGURE 1. **The 3-tier main framework of the proposed method: The first tier depicts the data acquisition and data pre-processing stage. The second tier involves ConvLSTM classification of objects, while the third tier integrates the YOLOv8l localizer for object detection.**

static regions in the foreground, identifying object types, and validating thresholds. Based on the outcomes of these earlier steps, if an object is determined to be unattended, the system triggers an alarm to alert the relevant authorities. Mahalingam and Subramoniam et al. [28] presented an effective method for tracking and detecting objects in videos that is divided into three separate stages: tracking, evaluating, and detecting. The MoAG model (Mixture of Adaptive Gaussian) was introduced to improve the efficiency of foreground segmentation during the detection phase, which includes noise reduction and foreground segmentation. Din et al. [29] have introduced a framework designed for the detection of abandoned luggage within public areas. The framework begins by utilizing the initial frames to model the background scene. To detect and track moving objects, including both the luggage owner and the luggage itself, the framework employs a model. Significantly, this method sustains its effectiveness in the face of challenges like affine distortion, noise, and occlusion. Moreover, for establishing the history of luggage, the model under consideration records the positional history of mobile objects and utilizes frame differencing as a fundamental technique. Hassan et al. [30] presented a real-time system to detect and track moving objects that become stationary in restricted zone a pixel classification method based on segmentation history image is used to identify stationary objects. Park et al. [19] introduced an algorithm specifically crafted to reliably detect abandoned objects, even when faced with changes in illumination. This algorithm demonstrates the ability to quickly adjust to a range of illumination variations.

It includes a presence authentication mechanism relying on the largest contour, allowing accurate tracking of target objects and the identification of abandonment, regardless of foreground information presence and the impact of illumination variations. Park et al. [31] presented a novel algorithm that accurately differentiates between stolen items, ghost regions, and abandoned objects by fusing conventional image processing methods with artificial intelligence technologies. This method uses two main strategies: segmenting objects using CNN-featured mask regions (Mask R-CNN) to provide comprehensive object mask information, and using a dual background model to detect possible stationary objects. Lwin and Tun [32] proposed a method using YOLOv4 to identify abandoned objects in video surveillance. This study developed a framework for detecting abandoned items by expanding the capabilities of YOLOv4. To support this research, a self-assembled dataset was used for training. The neural network was trained to predict six specific objects, including people, backpacks, handbags, books, hats and backpacks. Wahyono [33] introduced a dual a Gaussian mixture model-based

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

cumulative dual foreground difference is for stationary objects detection. An SVM-based classifier is then integrated to verify the region candidates whether they are vehicle, human or other objects. Palivela and Ramachandran [34] introduced a system for abandoned object detection, employing a hash-based approach and incorporating an SVM classifier. Their main area of interest is the identification of abandoned or unattended objects in indoor and outdoor environments. This is accomplished by taking video frames, extracting their features, and using machine learning algorithms to analyse them. Using hash value descriptors from the training phase, the performance of a binary SVM classifier is assessed, and a confusion matrix is used to compare the results with those from other classifiers.

Samaila et al. [35] have introduced a real-time vision-based abandoned object detection system. This system utilizes the Gaussian Mixture Model (GMM) and is capable of detecting objects as small as a teacup. It outperforms the Self-organizing Background Subtraction (SOBS) technique in handling background obstacles. Additionally, this system can classify abandoned items into two categories: non-human and human, a capability not found in other existing techniques. Palanisamy [36] proposed a mathematical technique called running average for video sequences of traffic taken from a static camera the background image is a static image that represents the scene without any moving objects. Chen et al. [37] have developed a systematic model pruning approach that evaluates the balance between accuracy and efficiency across diverse structured model pruning techniques and datasets, including CIFAR-10 and ImageNet. They used the VGG-16 model on Tensor Processing Units (TPUs) as a representative example. Additionally, they have introduced a structured model pruning package for TensorFlow2, allowing for in-place modification of models to evaluate their real-world performance. In their work, Palivela and Ramachandran [34] introduced a hash-based approach for abandoned object detection using an SVM classifier. They evaluated its performance in identifying and classifying unattended objects, comparing it to other classifiers through a confusion matrix. Table 1 provides a concise summary of key literature, offering insights into the main findings and methodologies employed by various studies in the field.

TABLE 1 KEY FINDINGS AND METHODOLOGIES FROM RELEVANT STUDIES IN THE FIELD

Author	Methodology	Main Findings
Wahyono [33]	- Dual background modeling - SVM classifier	- Detects and tracks stationary objects.
Hassan [30]	- Pixel classification - Adaptive tracking	- Real-time system achieving high detection success rate for stationary objects.
Palanisamy [36]	- Background modeling	- Detects stationary foreground objects in traffic videos based on their stationary time.
Park et al [19]	- Dual Background model - Mask R-CNN segmentation	- Presence authentication mechanism for accurate tracking and identification of abandonment.
Lwin and Tun [32]	- YOLOv4	- Framework for detecting abandoned objects, extending YOLOv4 capabilities. Trained on a self-assembled dataset predicting six specific object classes.
Palivela and Ramachandran [34]	- Hash-based approach - SVM classifier	- Detection of unattended items in public and indoor spaces. Evaluation using hash value descriptors and comparison with other classifiers.
Samaila et al. [35]	- Gaussian Mixture Model (GMM)	- Real-time vision-based abandoned object detection, capable of classifying abandoned items into non-human and human categories.

III. Proposed method

The proposed method comprises of three components: (1) an enhanced pre-processing step that enhances data quality through the use of advanced and refined techniques; (2) a ConvLSTM layer that captures both temporal and spatial information from the frames; and (3) the state-of-the-art YOLOv8, which identifies stationary abandoned objects within the frames. The complete workflow of the proposed method is depicted in Figure (1). The proposed method for abandoned object detection that can handle both suspicious and non- suspicious scenes. Firstly, SCM is based on a sequential model that can analyze the input image and

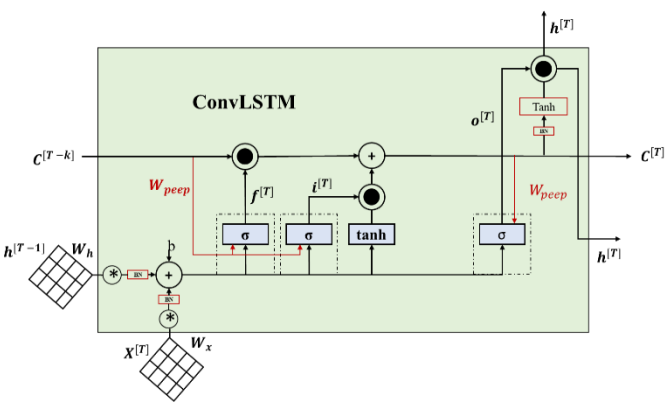


FIGURE 2. ConvLSTM structure

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

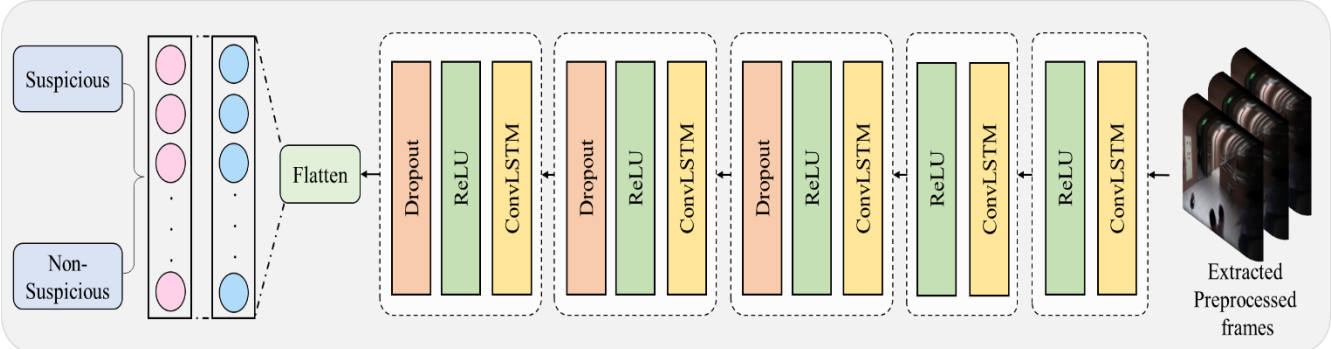


FIGURE 3. Propose ConvLSTM model architecture featuring five ConvLSTM layers, two dense layers with 1000 units each, and final classifier layer with one unit for classifying suspicious and non-suspicious objects

classify it as suspicious or non- suspicious scene. Secondly, ODM is based on the YOLOv8 architecture, which can locate and classify various types of objects in the input image. If the scene is classified as suspicious, the ODM will detect the objects that may be abandoned, such as luggage, backpacks, and boxes. Conversely if the scene is classified as non-suspicious, the object localization will not be attempted by ODM. The ODM can detect various objects that may include abandoned items, such as luggage, backpacks, boxes, etc. Our method can not only detect abandoned objects in real time, but also distinguish between different types of objects that may have different levels of risk or importance.

A. DATASETS DESCRIPTION

The dataset is a critical element for assessing the performance of any system. Evaluating the proposed algorithm using a well-established dataset presents a notable challenge in the domain of visual surveillance systems. In recent years, the availability of standard datasets for abandoned object detection has been limited.

1) PETS 2006 DATASET

PETS 2006 is a publically available dataset for abandoned object detection which we have used for our experiments. The dataset consists of 7 videos with 25 Frame Per Second (FPS) and standard resolution of 768x576 for evaluating the performance of object detection and tracking systems. The videos show different scenarios of left-luggage events in an outdoor parking lot, captured from multiple cameras and different angels. The dataset videos are annotated with bounding box and event type for abandoned object detection, such as left luggage, removed object, or vehicle movement.

2) ABODA DATASET

ABODA is a public dataset for abandoned object detection. The dataset consists of 11 video sequences captured from different CCTV footage, showing various rea-application scenarios that are challenging for abandoned object detection, such as crowded scenes, lighting changes, night-time detection, and indoor and outdoor environments. The videos are annotated with bounding boxes and ownership information of the abandoned object, indicating whether they belong to a person who is still present in the scene or not.

B. DATA PREPROCESSING

Pre-processing is a vital and indispensable step for achieving better performance in modeling. This process encompasses all the steps that enhance the quality of data, enabling the model to extract and interpret the data effectively. Consequently, the foundation of superior model performance is the pre-processing steps applied to the data before model feeding.

3) FRAMES EXTRACTION

Our data preprocessing steps involve frame extraction from and convert it into a batch of frames for models. We extract 15 frames per second from the video and discard irrelevant frames that have no significance for the model. This reduces the computational burden and likelihood of overfitting. We specifically retain the frames with totally distinct features relevant to the model, discarding irrelevant frames.

4) FRAME NORMALIZATION

The pre-processing steps involve the normalization of the characteristics of individual frames, this ensures consistency and comparability. Normalization frames can enhance the performance and reliability of the model. The formula of normalization is represented in equation (1):

$$\boldsymbol{I_{normalized}(x,y)} = \frac{\boldsymbol{I(x,y) - I_{min}}}{\boldsymbol{I_{max} - I_{min}}} \quad (1)$$

Where I(x,y) represent the intensity (brightness) value of the pixel. I_{min} is the minimum intensity value found in the frame. I_{max} is the maximum intensity value found in the frame.

5) FRAME CROPPING:

In the pre-processing steps, we cropped the frame to size 512,512 size before feeding it to the model. This allowed us to extract specific regions of interest from a larger frame, which is beneficial for various reasons. Firstly, cropping helps the model to focus on key details, enabling it to concentrate its attention on the most relevant information for the task at hand. Secondly, it reduces the data size, making it more manageable for subsequent processing and lower computational requirements. The mathematical form of the cropping can be represented as in the equation (2).

$$C = I\,[x_1 : x_2, y_1 : y_2] \quad (2)$$

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

Where I [x_1:x_2, y_1: y_2] represents a subarray or sub region of the original image or frame I. x1 and 1y1 specify the starting coordinates (top-left corner) of the crop region. x_2 and y_2 specify the ending coordinates (bottom-right corner) of the crop region.

6) FRAME AUGMENTATION:

In our preprocessing data, we augmented frames by applying various methods such as rotation, scaling, translation, flipping, and brightness adjustment. The process helps the model become resilient to variation in real-world scenarios such as illumination changes in viewpoint and object orientations. Moreover, frame augmentation can help in overcoming overfitting by expanding the volumetric dataset, thereby improving the model's ability to generalize to unseen data. The general formula of augmentation is given in equation (3).

$$A(I) = T(I, p_1, p_2, \dots, p_n) \quad (3)$$

Where A (I) represents the augmented version of the input data I. T denotes a data transformation function. p_1, $p_2 \dots$, p_n are parameters that control the specific augmentation techniques applied, such as rotation angles, scaling factors, translation distances, brightness adjustments, etc.

C. CONVLSTM ARCHITECTURE:

ConvLSTM is a convolutional neural network with an LSTM network. This is similar to LSTM with additional functionality of convolutional operation performed on the tensor, the structure of ConvLSTM is illustrated in figure 2. The network is suitable for sequential problems like videos where time is dependent. Spatial feature extraction in the model is accomplished through the utilization of convolutional layers. These layers apply filters to individual frames, allowing the capture of crucial spatial information such as object shapes and textures. On the other hand, the ConvLSTM component handles temporal feature extraction by maintaining hidden states, enabling it to grasp the evolution of video frames as they progress over time. This capability enables the model to understand motion, object interactions, and alterations in the video sequence. The key operation of ConvLSTM is given by the equation (4). Where * shows the convolution operator and ◦ shows the Hadamard metric.

$$\left\{\begin{array}{c} i_t = \sigma\left(W_{xi} * X_t + W_{hi} * H_{t-1} + W_{ci} \circ C_{t-1} + b_i\right), \\ f_t = \sigma\left(W_{xi} * X_t + W_{hf} * H_{t-1} + W_{cf} \circ C_{t-1} + b_f\right), \\ C_t = f_t C_{t-1} + i_t \tanh\left(W_{xc} * X_t + W_{hc} * H_{t-1} + b_f\right), \\ O_t = \sigma\left(W_{xo} * X_t + W_{ho} * H_t + W_{co} \circ C_{t-1} + b_o\right), \\ H_t = O_t \tanh(C_t) \end{array}\right\} \quad (4)$$

In this particular context, the variables are defined as follows: X_t represents the input to the cell, C_t corresponds to the cell's output, and the hidden state of the cell is denoted as H_t. Furthermore, i_t, f_t, and o_t are indicative of the input gate, and the sigmoidal function is represented by σ. The convolution kernels are denoted as W, as previously mentioned in the reference [20].

D. PROPOSED SEQUENTIAL CLASSIFIER MODEL:

In the proposed method we have harnessed a sequential model which extracts spatial-temporal features from the video. Recognizing the effectiveness of ConvLSTM, a widely acknowledged method for extracting features from sequential data, we integrated it into our framework to distinguish between static abandoned and non-abandoned objects. The architecture of our ConvLSTM model comprises five layers, with the initial layer serving as the input layer, accommodating sizes 15x512x512x3. Here '15' Signifies the sequence duration. While the subsequent dimensions correspond to the frame's spatial dimensions and color channels. The first layer incorporates 512, with a stride of 1, and padding set to 'same'. For the subsequent layers, the overall structure remains consistent, except for the number of filters. Specifically, the second employs 256 filters, the Third layer uses 128, the fourth layer employs 64, and the final layer involves 32 filters. After the third, fourth and last layers we employed dropout layers' rate of 0.5 values. Following this, a flattened layer and a flattening layer are applied to transform the 2D data into a 1D format, rendering it suitable for prediction. Subsequently, two dense layers with 1000 neurons at each, are introduced. Finally, the classification layer classifies static objects within the frames Table 2 presents the hyper-parameters employed in configuration of the proposed classifier model. Throughout the training, the dataset is divided into training, validation, and test sets with a distribution ratio of 70:20:10. Upon generating predictions, if the model's confidence surpasses a predefined threshold, the output is directed to a subsequent static localization model. This model effectively pinpoints the location of static objects within the frames enhancing the precision of the proposed method. Figure 3 illustrates the proposed classifier model architecture.

TABLE 2 HYPER-PARAMETERS OF THE PROPOSED CLASSIFER MODEL

Hyperparametre	Setting
Learning Rate	0.001
Optimizer	Batch Gradient Descent
Output layer (Classifier) activation function	Sigmoid
Input and headen layer activation function	ReLU
Number of layers	7
ConLSTM Layers	5
Dense Layers	2
Dropout rate	0.5

E. Proposed Yolov8 localizer model:

YOLOv8 is the most recent version of the YOLO object detection model. While maintaining the same foundational architecture as its predecessors, YOLOv8 incorporates several notable enhancements over prior versions. These include the adoption of an advanced neural network structure that leverages both Feature Pyramid Network (FPN) and Path Aggregation Network (PAN), alongside the introduction of an

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

improved labeling tool designed to streamline the annotation process. The labeling tool encompasses a range of valuable functionalities, including automated labeling, convenient labeling shortcuts, and the ability to customize hotkeys. The synergy of these capabilities significantly simplifies the process of annotating images for model training purposes. Additionally, it's worth noting that YOLOv8 has several versions, such as YOLOv8-S, YOLOv8-M, YOLOv8-L, and YOLOv8-XL. The Feature Pyramid Network (FPN) operates by progressively decreasing the spatial resolution of the input image while concurrently augmenting the quantity of feature channels. This process leads to the generation of feature maps with the capacity to identify objects at varying scales and resolutions. In contrast, the Path Aggregation Network (PAN) architecture integrates features from diverse network levels by means of skip connections. Through this approach, the network becomes more adept at capturing features across various scales and resolutions. This capability is of utmost importance for achieving precise object detection, particularly when dealing with objects of diverse sizes and shapes [18].

1) YOLOV8 LOCALIZER MODEL SPECIFICATIONS:

In the concluding phase of our proposed methodology that detects the static abandoned with higher precision. We deliberately selected Yolov8 as the proposed model for static abandoned object detection, under the premise that exhibits the highest probability for detecting stationary objects. Yolov8 is the latest state-of-the-art method known for its higher Mean Average precision (mAPs) and having lower inference speed. The model has been meticulously trained on one of the most well-known datasets, the COCO dataset. Our research also entailed a comprehensive evaluation of various state-of-the-art object detection models including Faster-RCNN [38], Fast-RCNN, SSD [39]. Among these all yolov8 series outperformed it counterparts, consistently achieving either a higher precision or faster inference time. In our proposed method, YOLOv8 utilizes CSPDarknet53 as its backbone. CSPDarknet53 is a deep neural network that excels in extracting features at various resolutions or scales by gradually reducing the size of the input image. The feature maps generated at different resolutions hold essential information about objects present in the image at various scales, offering varying levels of detail and abstraction. Our approach harness Yolov8's capability to leverage these diverse features maps at different features map at different scales to gain insight the object morphology and texture of objects, thereby enhancing precision in detection of static abandoned detection objects. Yolov8 backbone consists of four sections, each commencing with single convolution followed by a c2f module [40]. Notably, our methodology leverages C2F module introduced CSPDarknet53. The module incorporates splits where one branch traverses through a bottleneck module characterized by Two 3x3 convolution with residual connections. The output of bottleneck module undergoes further splitting, occurring N times, with N corresponding to the Yolov8 model size. These splits are subsequently concatenated and channeled through a final convolution layer, which serves as the layer responsible for activating the network. This integrated architecture enhances our approach's capability to detect static abandoned objects effectively. The activation map associated with the shallowest c2f module reveals prominent activations corresponding to the object of boards. This module specializes in detecting small objects and identifying their respective classes. Moving to the second activation, it plays a crucial role in determining the presence of static abandoned objects. As we delved deeper into the model, the third activation started capturing intricate texture associated static abandoned objects. Ultimately, the model's final C2F module activates to capture exceptionally fine-grained details and outlines within the image under consideration.

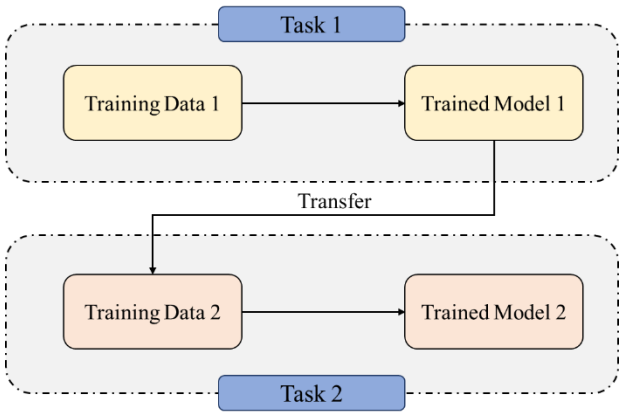


FIGURE 4. Transfer learning mechanism

2) LOCALIZER OPTIMIZATION USING TRANSFER LEARNING:

Yolov8 originally pre-trained on the COCO dataset which encompasses a wide array of object classes. However, our objective pertains specifically to detecting static objects such as bags. In order to optimize the model's performance for this specialized task, it must be tailored accordingly. Training for the specific from scratch would be prohibitively expensive and need more specific dataset. Therefore, we adapted Transfer leaning [41] technique to leverage the knowledge acquired during pretraining on COCO. This enables us to adapt model's weights and features to better suit the detection of bags while utilizing a limited dataset. Transfer learning approach is visualized in figure 4.

F. Evaluation metrics:

The performance of the proposed methodology is evaluated using specific performance measures including accuracy, precision, and recall.

1) Accuracy:

Accuracy is a metric commonly employed to provide an overall assessment of a model's performance across all classes. This metric is particularly valuable when all classes hold equal significance, quantifying the correctness of predictions by dividing the number of accurate predictions by the total number of predictions, equation (5) demonstrates accuracy.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (5)$$

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

2) Precision:

Precision is determined by dividing the number of Positive samples correctly classified by the total count of samples classified as Positive, whether they were classified correctly or not. It serves as an indicator of the model's accuracy in classifying samples as positive, specifically measuring its ability to correctly identify positive instances. Precision is presented by Equation (6).

$$Precision = \frac{TP}{TP+FP} \tag{6}$$

3) Recall:

Recall is computed by dividing the number of correctly classified Positive samples by the total count of Positive samples. It quantifies the model's capability to identify Positive samples accurately, essentially gauging its sensitivity in detecting such instances. A higher recall value signifies a greater ability to detect Positive samples within the dataset. Recall is illustrated in equation (7).

$$Recall = \frac{TP}{TP+FN} \tag{7}$$

Where True Positive (TP): The model correctly identifies something as positive, and it is indeed positive. True Negative (TN): The model correctly recognizes something as negative, and it is indeed negative. False Positive (FP): The model mistakenly identifies something as positive when it's actually negative. False Negative (FN): The model erroneously identifies something as negative when it's actually positive.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. EXPERIMENTAL SETUP

In this section, we present a comprehensive overview of the experimental setup of the proposed methodology. The system implementation was executed using Python 3.10.4, PyTorch. 1.12.1, and CUDA 11.7, with both training and inference processes conducted on a powerful 12GB NVIDIA GeForce RTX 3090 GPU. For our primary model, convLSTM, we conducted training over a span of 100 epochs. The weights of the model were updated during backward propagation using Adam optimizer. We configured a batch size of 64 to balance training efficiency and GPU memory usage effectively. Our proposed model tailored specifically for the task of abandoned detection comprises two distinct categories: abandoned and non-abandoned objects. To optimize for the binary classification, we employed binary Cross-Entropy losses during training. It is important to note that yolov8 served as the foundation for our object detection task. The model underwent 100 epochs for training. We fine-tuned it using training learning for our specific use case. This transfer learning approach allowed us to adapt Yolov8's pre-trained weights to intricacies of abandoned object localization within limited dataset. In the term of data preparation, we meticulously annotated the proposed dataset for model to localize the static object. We annotated 500 images for the model from the different proposed datasets. The annotation process involved defining bounding boxes around the objects of interest, providing essential training data for our models. Our dataset, sourced from diverse environments and scenarios, features variations in lighting conditions, backgrounds, and object poses, reflecting the real-world challenges of abandoned object detection. Ethical considerations guided our data collection, and privacy and bias mitigation measures were considered.

B. Results and Discussion

In this section, we embark on an empirical evaluation proposed model. This evaluation is structured into three main components. Firstly, in the Architectural Variations Analysis section, we delve into the inner model comparison, where we assess the model's performance using various deep learning architectures for temporal features extraction. Secondly, is the object detection model evaluation where we conduct a detailed analysis of the object detection model integrated into our proposed framework, aiming to identify strengths and potential areas for improvement Finally, State-of-the-art model in which, we compare the effectiveness of our proposed model with other state-of-the-art models in the field. This comparative analysis offers valuable insights into the model's performance and its standing with boarder research landscape.

1) Architectural Variation Analysis of Classifier

To validate the effectiveness of our proposed model in comparison to various sequential models, we conducted a series of empirical experiments. Upon acquiring visual data, we directed this data sequence to sequential models designed to extract temporal information, a crucial element for precise predictions. This approach capitalizes on the understanding that the quality of features extracted from the frames significantly influences prediction accuracy. During the experiment, our goal was to identify optimal model term of extracting these enhanced features, which are pivotal for precise temporal predictions. To achieve this, we employed a range of diverse deep learning models for features including the Gated Recurrent Unit (GRU) model, Recurrent Neural Network (RNN), Vanilla Long-Short Term Memory (LSTM) model, and our proposed model. These experiments helped to seek to discern the model that outperforms others in the term of enhancing features critical for precise temporal predictions. We chose the best sequential model based on its superior precision, with the parameter determining our selection. The outcome of different model along with proposed is illustrated in the Table 3.

TABLE 3 SEQUENTIAL CLASSIFIER MODELS PERFORMANCE

Model	Accuracy (%)	F1-Score (%)
GRU	84.67	80.03
RNN	89.01	88.43
LSTM	95.56	90.40
Bi-Directional LSTM	98.88	95.82
ConvLSTM	**99.20**	**99.05**

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

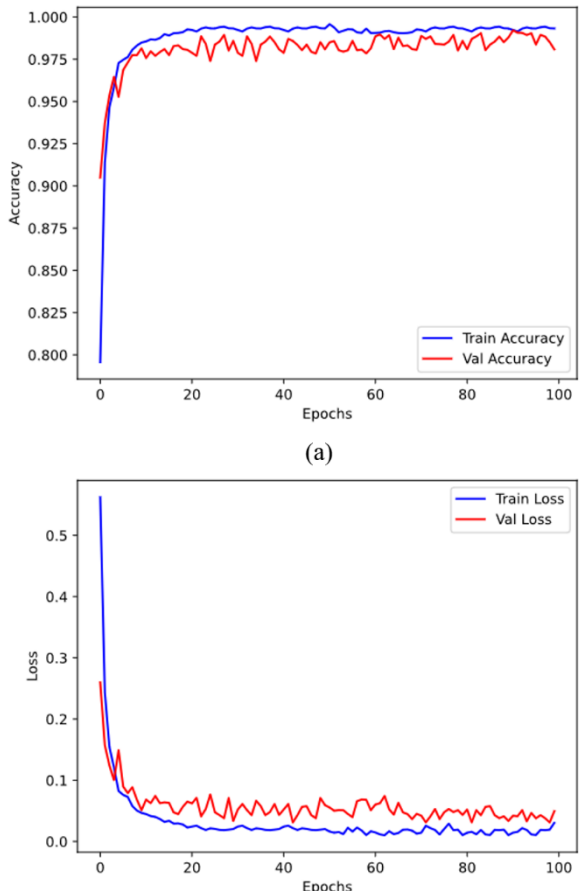


FIGURE 6. Proposed Classifier model learning graphs. (a) Accuracy of the model on the training and validation sets. (b) Loss of the model on training and validation sets.

The training phase is divided into two segments: training phase for 50 and 100 epochs respectively. In the first we trained these models for 50 epochs. We compared five sequential models for sequence learning namely (GRU, RNN, LSTM, Bi-Directional LSTM, and ConvLSTM). The vanilla GRU model suffered from overfitting, the RNN from vanishing gradients, and the vanilla LSTM from poor feature extraction. The ConvLSTM outperformed the other models, achieving the highest training and validation accuracy of 80% in the first training phase. In the second phase, we repeated experiments to train these models for 100 epochs. Firstly, the Vanilla GRU model underwent training, during training we kept the sequence of 8 in the data to capture relation better. The model performance didn't show any significant improvement, rather the model was overfitting on the data. The result was slightly improved but not promising. The model achieved the training and validation accuracy of 55% and 26%. Secondly, in the training phase of RNN, the sequence for the model was 8 to avoid vanishing gradient problem. During training, we evaluated the model performance was poor on both training and validation data. The model showed a higher underfitting problem. The longer training could not improve the model performance rather than a tinny point improvement. Empirically, the training and validation accuracy of the model was 40% and 30%. Thirdly, Vanilla LSTM model underwent training with sequence of 12. Throughout the training the model performance fluctuated, the model showed smooth performance. Consequently, in the end the model could not perform better. Resultantly, the model achieved the training and validation accuracy of 52% and 41%. Finally, we trained the convLSTM with 15 sequences in the data. The model showed significant performance throughout the training. The model generalization was remarkably improved as we kept the training. The model outperformed all the remaining models. Empirically we found, the model achieved the training and validation accuracy of 99.20% and 99.50%, respectively, with an impressive F1-score of 99.05%. Based on these evaluated metrics led to the selection of our model for sequential feature extraction.

Figure 5 shows the training and validation accuracy and loss of the proposed model throughout the training. The x-axis of the graphs shows the Number of epochs while the y-axis shows the performance of the model. The Figure 5(a) indicates at the beginning of training the model started with higher training and validation accuracy. Throughout the training the model showed better generalization however, at the epoch from 20 to 30 model showed some fluctuation but onward that there is significant improvement. The graphs showed the model was capturing the relation very smoothly. The fluctuation portion of the model showed the adjustment for learning unseen data, in addition further training could cause overfitting. Resultantly, the proposed model achieved the highest accuracy among other different sequential models. On the other hand, Figure 5(b) showed the training and validation loss of the proposed model. At the beginning of training, the drastic deceased in the model showed better generalization, throughout the training both losses were smoothly decreasing. The final portion of training showed some overfitting therefore, we stopped the model on 100 epochs. The training and validation loss of the model was 0.01% and 0.02% respectively. Figure 6 shows the confusion matrix of the proposed model, this shows the miss prediction value and the true value for each class. It can be seen from the figure, that abandoned class has lower accuracy compared to the non-abandoned class, resulting in the model achieving overall accuracy of 99.20%.

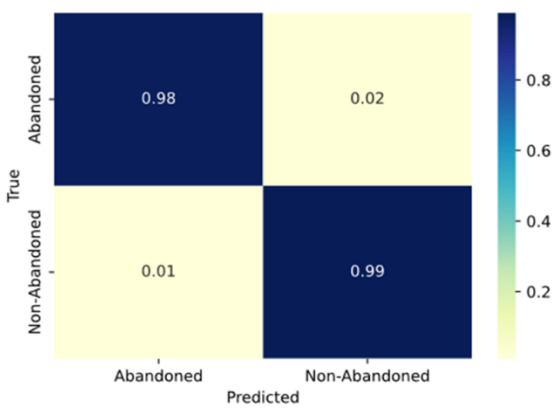


FIGURE 5. Proposed classifier model confusion matrix

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

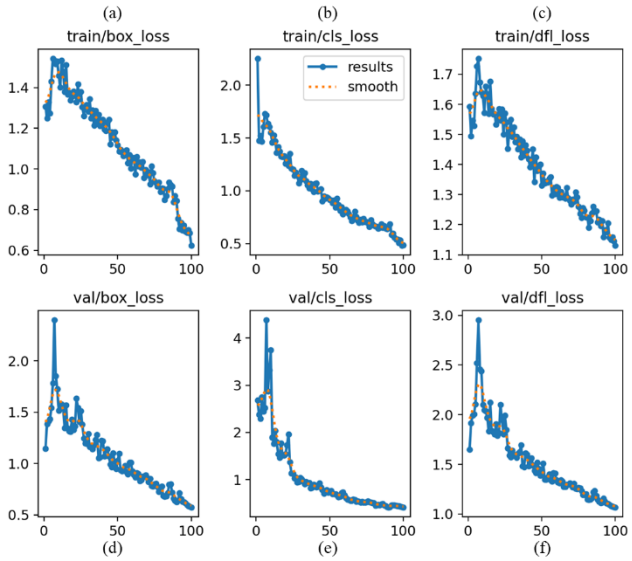


FIGURE 7. Proposed Localizer model training and validation loss graphs. (a) Training box loss. (b) Training classification loss. (c) Training Distribution focal loss. (d) Validation box loss. (e) Validation classification loss. (f) Validation distribution focal loss.

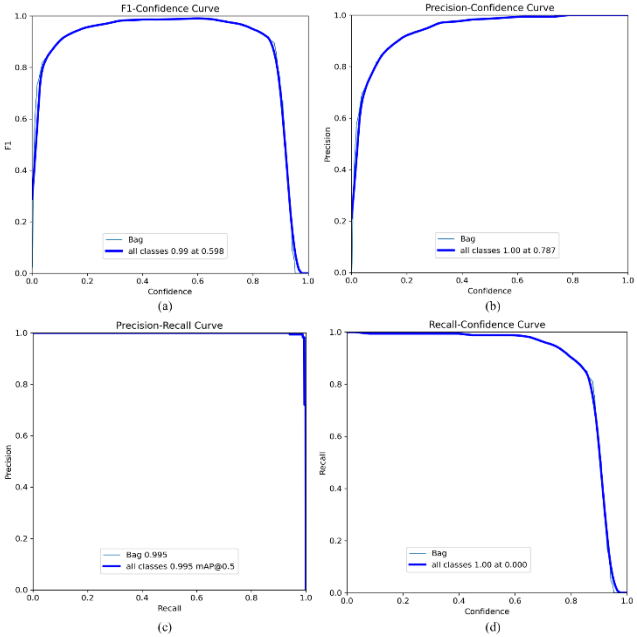


FIGURE 8. Other evaluation graphs of the proposed localizer model. (a) F1-score confidence curve. (b) Precision confidence curve. (c) Precision recall curve. (d) Recall confidence curve.

2) Architectural Variation Analysis of localizer

In this section, we present our research in three parts. Firstly, we discuss the selection of a robust YOLOv8 variant for abandoned object detection. Secondly, we evaluate our proposed localizer model against various YOLOv8 variants, including YOLOv8-n, YOLOv8-m, YOLOv8-l, YOLOv8-x, and YOLOv8-s. Our experiments involve test images of abandoned objects, and the results are summarized in Table 4, covering precision, recall, F1-score, mAP50, and mAP50-95 metrics. Notably, our proposed YOLOv8l-seg stands out with the highest precision 99.7% and recall 99.5% in abandoned object detection. The evaluation highlights the influence of model size and dataset characteristics on performance, with denser models showing fewer promising results. Specifically, YOLOv8n-seg exhibits the lowest precision 92.4% and recall 89.1%, YOLOv8x-seg slightly lower precision 95.6% and recall 93.2, YOLOv8m-seg demonstrates precision 98.2% and recall 97.7%, and YOLO8s-seg secures a lower precision score of 94.1% and recall of 96.2%.

The proposed approach for identifying and categorizing stationary objects demonstrates its versatility and effectiveness across a spectrum of scenarios, including public transportation hubs, commercial centers, urban streets, public events, smart city infrastructure, residential areas, critical infrastructure sites, and outdoor parks, showcasing its robust applicability in diverse real-world environments, as depicted in Figure 9. This approach significantly enhances precision in localization, a crucial aspect for subsequent classification tasks. Our precision-recall confidence curve achieves an impressive 99.0% mAP for all classes, illustrated in Figure 8 (c). We conducted a comparative analysis with state-of-the-art models and refine our method for optimal abandoned object detection performance.

The proposed model underwent a comprehensive evaluation during training and validation, assessing key metrics as shown in Figure 7. Notably, the training box loss decreased to 0.06, indicating high confidence in predictions. Validation results, depicted in Figure 7 (d, e, f), revealed significant reductions in box loss (0.5), class loss (0.01), and df1 loss (0.10), showcasing the model's outstanding performance.

Figure 8 presents evaluations using precision-recall curves, precision-confidence curves, recall-confidence curves, and f1-confidence curves for abandoned object detection. The precision-recall curve consistently yields high values 99.0%, indicating robust performance in segmenting abandoned objects. The precision-confidence curve affirms the model's accurate identification, while the recall-confidence curve demonstrates correct identification of all positive instances. The F1-confidence curve shows a balanced trade-off between recall and precision scores, with an F1-score of 1.00%, emphasizing the model's superior performance in accurately segmenting various abandoned objects.

TABLE 4 YOLOV8 MODELS PERFORMANCE

Model Name	Precision	Recall	F1-Score	mAP50	Map50-95
Yolov8n	0.924	0.891	0.843	0.815	0.799
Yolov8m	0.98	0.977	0.923	0.901	0.852
Yolov8x	0.956	0.932	0.931	0.915	0.932
Yolov8s	0.941	0.962	0.900	0.888	0.799
Yolov8l	**0.997**	**0.995**	**0.991**	**0.965**	**0.953**

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

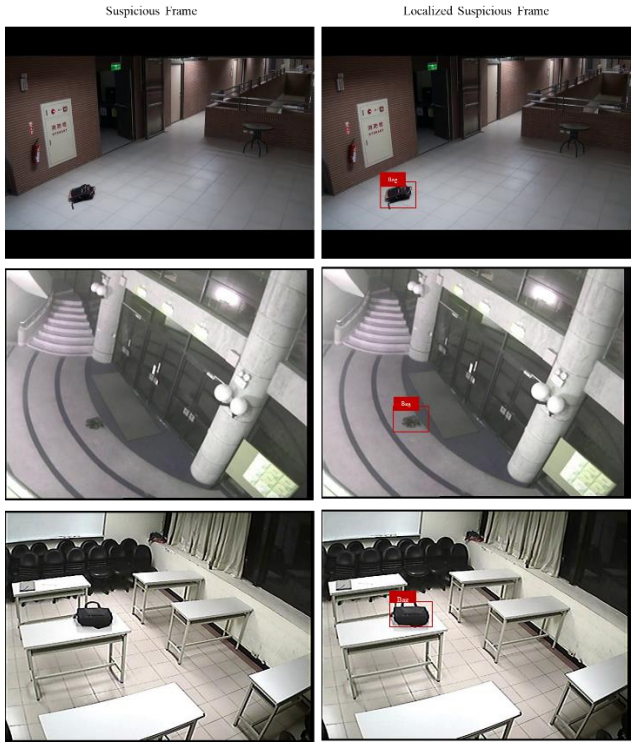


FIGURE 9. Suspicious object localization results on ABODA dataset using YOLOv8l localizer

3) Computational Complexity analysis

This section provides a detailed analysis of the computational complexity of our proposed method for abandoned object detection. We have consider various factors, including number of parameters, memory usage, training time, and inference speed. Table 5 presents a comprehensive breakdown of the computational complexity analysis for classifier models. Inference time for the classifiers were determined based on a sequence length of 10. Notably, our proposed ConvLSTM model exhibits a lightweight architecture, resulting in faster inference time compared to other models in evaluation.

TABLE 5 COMPUTATIONAL COMPLEXITY ANALYSIS OF CLASSIFIER MODELS

Model	Parameters	Memory Usage	Training Time	Inference Speed (sec)
GRU	243563	18.53 MB	18 min	1.01 sec
RNN	536540	34.00 MB	29 min	1.30 sec
LSTM	634234	39.00 MB	43 min	1.00 sec
Bi-Directional LSTM	300123	20.00 MB	23 min	0.50 sec
ConvLSTM	25839	8.05 MB	10 min	0.30 sec

Table 6 presents complexity analysis of various YOLOv8 models. Our proposed YOLOv8l model stands out with 102.2 million parameters, consuming 408.8 MB of memory and requiring 30 minutes for training, and achieving an inference speed of 29 milliseconds.

TABLE 6 COMPUTATIONAL COMPLEXITY ANALYSIS OF LOCALIZER MODELS

Model	Parameters (million)	Memory Usage	Training Time	Inference Speed (millisecond)
YOLOv8n	4.3M	4.3 MB	15 min	10 ms
YOLOv8s	12.3M	12.3 MB	20 min	15 ms
YOLOv8m	35.6M	35.6 MB	23 min	20 ms
YOLOv8l	102.2M	408.8 MB	30 min	28 ms
YOLOv8x	235.9M	943.6 MB	35 min	31 ms

4) Comparative Analysis of the proposed method with State-of-the-art models:

This section provides a comprehensive breakdown of the performance analysis and a comparative assessment of both the established and our proposed methods. An internal localizer comparison, specifically focusing on YOLOv8, has been carried out to enhance the evaluation. Table 2 illustrates the performance of various localizers, where YOLOv8l emerges as the top performer with a precision score of 99.7%, recall score of 99.5%, and an impressive F1-Score of 99.1%.
For proposed method and state of the art methods comparison we closely scrutinize key performance metrics, including Accuracy, Precision, and Recall. The proposed model underwent a comprehensive evaluation, comparing it to pre-existing methods. Empirically, the results demonstrated that the proposed model significantly outperformed all existing state-of-the-art models by achieving a substantial increase in accuracy. The detailed comparison findings are outlined in Table 7.

TABLE 7 COMPARISON OF THE PROPOSED LOCALIZER WITH SOTA

Author	Accuracy (%)	Precision (%)	Recall (%)
Mahalingam and M. Subramoniam [26]	65.00	99.00	99.00
Lwin and Tun. [19]	84.55	98.80	97.70
Palivela and Ramachandran [31]	95.15	98.80	98.40
Dr. Kiruthig and Yuvaraj [32]	83	79.00	86.00
Teja, YD [10]	99	91.00	88.00
Proposed Method	**99.70**	**99.98**	**99.50**

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

Figure 10 presents a graphical comparison of the performance metrics, including Accuracy, Precision and Recall between the existing method and proposed one. Significantly, the proposed method distinctly outperforms the mentioned existing methods by a substantial margin.

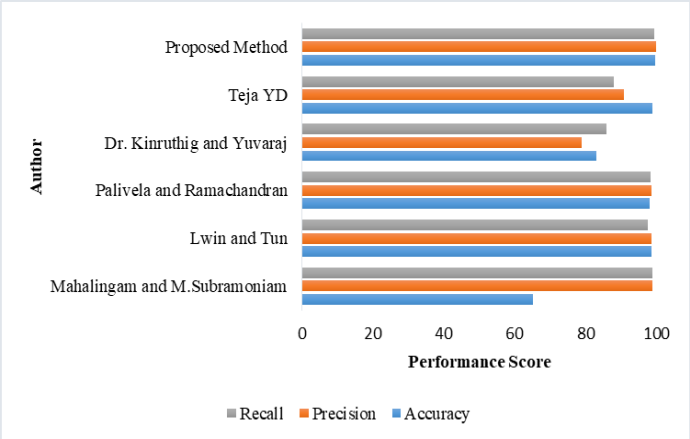


FIGURE 10. Proposed method and SOTA comparison graphical representation

IX. CONCLUSION

In the realm of video surveillance, a significant yet challenging focus lies on the realm of automatic event detection. Particularly, the detection of abandoned objects (AOD) has garnered substantial attention in recent times, as it plays a critical role in enhancing security in both public and private domains. The global concerns of security and terrorism have escalated to unprecedented levels over the past years, with terrorist attacks claiming innocent lives, often striking crowded locations such as markets, transportation hubs, and airports. To address these pressing security issues effectively, the deployment of automated surveillance technologies in public spaces has become increasingly imperative. In conclusion, our proposed model incorporates a sequential analysis, operating on sequences of 15 frames for initial object detection. This sequential approach facilitates a thorough exploration of temporal patterns and characteristics. Subsequently, the processed data seamlessly advances to the YOLOv8 model, renowned for its exceptional object localization capabilities. By merging the temporal insights derived from the sequential model with YOLOv8's precision in pinpointing object locations, our approach offers a comprehensive and effective solution for object detection and localization tasks. This integration represents a significant stride in enhancing the field of security and surveillance.

REFERENCES

[1] S. Kalli, T. Suresh, A. Prasanth, T. Muthumanickam, and K. Mohanram, "An effective motion object detection using adaptive background modeling mechanism in video surveillance system," *Journal of Intelligent & Fuzzy Systems,* vol. 41, pp. 1777-1789, 2021.

[2] M. Elhoseny, "Multi-object detection and tracking (MODT) machine learning model for real-time video surveillance systems," *Circuits, Systems, and Signal Processing,* vol. 39, pp. 611-630, 2020.

[3] N. Bird, S. Atev, N. Caramelli, R. Martin, O. Masoud, and N. Papanikolopoulos, "Real time, online detection of abandoned objects in public areas," in *Proceedings 2006 IEEE International Conference on Robotics and Automation, 2006. ICRA 2006.*, 2006, pp. 3775-3780.

[4] K.-H. Jo, "Cumulative dual foreground differences for illegally parked vehicles detection," *IEEE Transactions on Industrial Informatics,* vol. 13, pp. 2464-2473, 2017.

[5] E. Luna, J. C. San Miguel, D. Ortego, and J. M. Martínez, "Abandoned object detection in video-surveillance: survey and comparison," *Sensors,* vol. 18, p. 4290, 2018.

[6] M. A. Mahale, H. Kulkarni, and P. Student, "Survey on Abandoned Object Detection in Surveillance Video," *Int. J. Eng. Sci. Comput,* vol. 7, pp. 15595-15599, 2017.

[7] S. Jha, C. Seo, E. Yang, and G. P. Joshi, "Real time object detection and trackingsystem for video surveillance system," *Multimedia Tools and Applications,* vol. 80, pp. 3981-3996, 2021.

[8] V. Akre, A. Rajan, J. Ahamed, A. Al Amri, and S. Al Daisi, "Smart digital marketing of financial services to millennial generation using emerging technological tools and buyer persona," in *2019 Sixth HCT Information Technology Trends (ITT)*, 2019, pp. 120-125.

[9] B. Qian, Z. Wen, J. Tang, Y. Yuan, A. Y. Zomaya, and R. Ranjan, "OsmoticGate: Adaptive Edge-Based Real-Time Video Analytics for the Internet of Things," *IEEE Transactions on Computers,* vol. 72, pp. 1178-1193, 2022.

[10] Y. Teja, "Static object detection for video surveillance," *Multimedia Tools and Applications,* pp. 1-13, 2023.

[11] S. Khan and L. AlSuwaidan, "Agricultural monitoring system in video surveillance object detection using feature extraction and classification by deep learning techniques," *Computers and Electrical Engineering,* vol. 102, p. 108201, 2022.

[12] D.-y. Ge, X.-f. Yao, W.-j. Xiang, and Y.-p. Chen, "Vehicle detection and tracking based on video image processing in intelligent transportation system," *Neural Computing and Applications,* vol. 35, pp. 2197-2209, 2023.

[13] A. Sathesh and Y. B. Hamdan, "Speedy detection module for abandoned belongings in airport using improved image processing technique," *J Trends Comput Sci Smart Technol,* vol. 3, p. 251, 2021.

[14] N. Dwivedi, D. K. Singh, and D. S. Kushwaha, "An approach for unattended object detection through contour formation using background subtraction," *Procedia Computer Science,* vol. 171, pp. 1979-1988, 2020.

[15] V. Indhuja, M. V. Reddy, N. Nikhitha, and P. Pramila, "Suspicious Activity Detection using LRCN," in *2023 5th International Conference on Smart Systems and Inventive Technology (ICSSIT)*, 2023, pp. 1463-1470.

[16] H. Su, W. Wang, and S. Wang, "A robust all-weather abandoned objects detection algorithm based on dual background and gradient operator," *Multimedia Tools and Applications,* pp. 1-23, 2023.

[17] N. Ta, H. Chen, Y. Lyu, X. Wang, Z. Shi, and Z. Liu, "A complementary and contrastive network for stimulus segmentation and generalization," *Image and Vision Computing,* vol. 135, p. 104694, 2023.

[18] J. Ju and J. Xing, "Moving object detection based on smoothing three frame difference method fused with RPCA," *Multimedia tools and applications,* vol. 78, pp. 29937-29951, 2019.

[19] H. Park, S. Park, and Y. Joo, "Robust detection of abandoned object for smart video surveillance in illumination changes," *Sensors,* vol. 19, p. 5114, 2019.

[20] H. Lee, J. Yoon, Y. Jeong, and K. Yi, "Moving object detection and tracking based on interaction of static obstacle map and geometric model-free approachfor urban autonomous driving," *IEEE Transactions on Intelligent Transportation Systems,* vol. 22, pp. 3275-3284, 2020.

[21] A. B. Mabrouk and E. Zagrouba, "Abnormal behavior recognition for intelligent video surveillance systems: A

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4

This article has been accepted for publication in IEEE Access. This is the author's version which has not been fully edited and
content may change prior to final publication. Citation information: DOI 10.1109/ACCESS.2024.3369233

review," *Expert Systems with Applications,* vol. 91, pp. 480-491, 2018.
[22] V. Tsakanikas and T. Dagiuklas, "Video surveillance systems-current status and future trends," *Computers & Electrical Engineering,* vol. 70, pp. 736-753, 2018.
[23] S. Ammar, T. Bouwmans, N. Zaghden, and M. Neji, "Moving objects segmentation based on deepsphere in video surveillance," in *Advances in Visual Computing: 14th International Symposium on Visual Computing, ISVC 2019, Lake Tahoe, NV, USA, October 7–9, 2019, Proceedings, Part II 14*, 2019, pp. 307-319.
[24] P. Grandhe, P. B. Dhanush, M. Mohammad, A. N. A. A. Lakshmi, and C. V. S. R. Kumar, "An Extensive Study on Unattended Object Detection in Video Surveillance," in *International Conference on Intelligent Sustainable Systems*, 2023, pp. 183-193.
[25] Q. Fan and S. Pankanti, "Modeling of temporarily static objects for robust abandoned object detection in urban surveillance," in *2011 8th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, 2011, pp. 36-41.
[26] E. Omrani, H. Mousazadeh, M. Omid, M. T. Masouleh, H. Jafarbiglu, Y. Salmani-Zakaria*, et al.*, "Dynamic and static object detection and tracking in an autonomous surface vehicle," *Ships and Offshore Structures,* vol. 15, pp. 711-721, 2020.
[27] P. Narwal and R. Mishra, "Real time system for unattended Baggag e detection," *Proceedings of the International Research Journal of Engineering an d Technology (IRJET),* vol. 6, 2019.
[28] T. Mahalingam and M. Subramoniam, "A robust single and multiple moving object detection, tracking and classification," *Applied Computing and Informatics,* vol. 17, pp. 2-18, 2021.
[29] M. Din, A. Bashir, A. Basit, and S. Lakho, "Abandoned object detection using frame differencing and background subtraction," *International Journal of Advanced Computer Science and Applications,* vol. 11, 2020.
[30] W. Hassan, P. Birch, B. Mitra, N. Bangalore, R. Young, and C. Chatwin, "Illumination invariant stationary object detection," *IET Computer Vision,* vol. 7, pp. 1-8, 2013.
[31] H. Park, S. Park, and Y. Joo, "Detection of abandoned and stolen objects based on dual background model and mask R-CNN," *IEEE Access,* vol. 8, pp. 80010-80019, 2020.
[32] S. P. Lwin and M. T. Tun, "Deep convonlutional neural network for abandoned object detection," *Int Res J Mod Eng Technol Sci,* vol. 4, pp. 1549-1553, 2022.
[33] Wahyono, R. Pulungan, and K.-H. Jo, "Stationary object detection for vision-based smart monitoring system," in *Intelligent Information and Database Systems: 10th Asian Conference, ACIIDS 2018, Dong Hoi City, Vietnam, March 19-21, 2018, Proceedings, Part II 10*, 2018, pp. 583-593.
[34] L. H. Palivela and S. Ramachandran, "An enhanced image hashing to detect unattended objects utilizing binary SVM classification," *Journal of Computational and Theoretical Nanoscience,* vol. 15, pp. 121-132, 2018.
[35] Y. Samaila, H. Rabiu, and I. Mustapha, "Real-time detection of abandoned object using centroid difference method," *Arid Zone Journal of Engineering, Technology and Environment,* vol. 16, pp. 48-57, 2020.
[36] H. Smitha and V. Palanisamy, "Detection of stationary foreground objects in region of interest from traffic video sequences," *International Journal of Computer Science Issues (IJCSI),* vol. 9, p. 194, 2012.
[37] K. Chen, K. Franko, and R. Sang, "Structured model pruning of convolutional networks on tensor processing units," *arXiv preprint arXiv:2107.04191,* 2021.
[38] B. Cheng, Y. Wei, H. Shi, R. Feris, J. Xiong, and T. Huang, "Revisiting rcnn: On awakening the classification power of faster rcnn," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 453-468.
[39] S. Zhai, D. Shang, S. Wang, and S. Dong, "DF-SSD: An improved SSD object detection algorithm based on DenseNet and feature fusion," *IEEE access,* vol. 8, pp. 24344-24357, 2020.
[40] D. Reis, J. Kupec, J. Hong, and A. Daoudi, "Real-Time Flying Object Detection with YOLOv8," *arXiv preprint arXiv:2305.09972,* 2023.
[41] F. Zhuang, Z. Qi, K. Duan, D. Xi, Y. Zhu, H. Zhu*, et al.*, "A comprehensive survey on transfer learning," *Proceedings of the IEEE,* vol. 109, pp. 43-76, 2020.



Arbab Muhammad Qasim is PhD Scholar in department of computer science at Islamia College University Peshawar, Pakistan since 2020. He received his master's degree in computer science from Iqra National University, Peshawar, Pakistan



Naveed Abbas received PhD degree in computer science from University Technology Malaysia, 2016. He is currently working Assistant Professor at department of computer science Islamia College University Peshawar, Pakistan



Amjid Ali received his BS degree in computer science from Islamia College University, Peshawar, Pakistan in 2021. He is currently working as assistant researcher at the same university.



Dr. Bandar Ali Al-Rami Al-Ghamdi, holds a PhD from Universite de Reims Champagne-Ardenne in 2015, Reims, France, the M.Sc. degree in Information Tech- nology from De Montfort University, Leicester, United Kingdom, in 2008 and the B.Cs. degree in Computer Sciences from the University of King Abdul-Aziz Univer- sity, Jeddah, Saudi Arabia, in 2003. Currently he is an assistant professor at Arab Open University, Riyadh, KSA. His research interests are Sensor Networks, Distributed Systems, eHealth Systems.

This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 License. For more information, see https://creativecommons.org/licenses/by-nc-nd/4